Exp No: 9	**Implementation of SUBNETTING in CISCO Packet Tracer Simulator**

AIM:

To design multiple subnets with a suitable number of hosts, assign static IP addresses, connect the subnets through a router in Cisco Packet Tracer, and observe packet transmission with and without the router.

ALGORITHM / PROCEDURE:

1. Take the network address 192.168.1.0/24 and decide the number of subnets needed (3 departments, so 4 subnets are created by borrowing 2 host bits).
2. The new mask is 255.255.255.192 (/26). Each subnet has 64 addresses, so 62 usable hosts.
3. Open Packet Tracer and place 1 router (2911), 3 switches (2960) and 6 PCs. Connect them with copper straight-through cables as in Fig 1.
4. Assign a static IP address, subnet mask and default gateway to every PC under *Desktop > IP Configuration*.
5. Configure the router interfaces: `interface gig0/0`, `ip address 192.168.1.1 255.255.255.192`, `no shutdown`
6. Repeat step 5 for Gig0/1 and Gig0/2 with the gateway address of subnets B and C.
7. From the PC command prompt use `ping` to test connectivity inside a subnet and across subnets.
8. Remove the router (delete the links to the router) and ping across the subnets again to see the difference.

Fig 1: Network topology with three subnets joined by a router

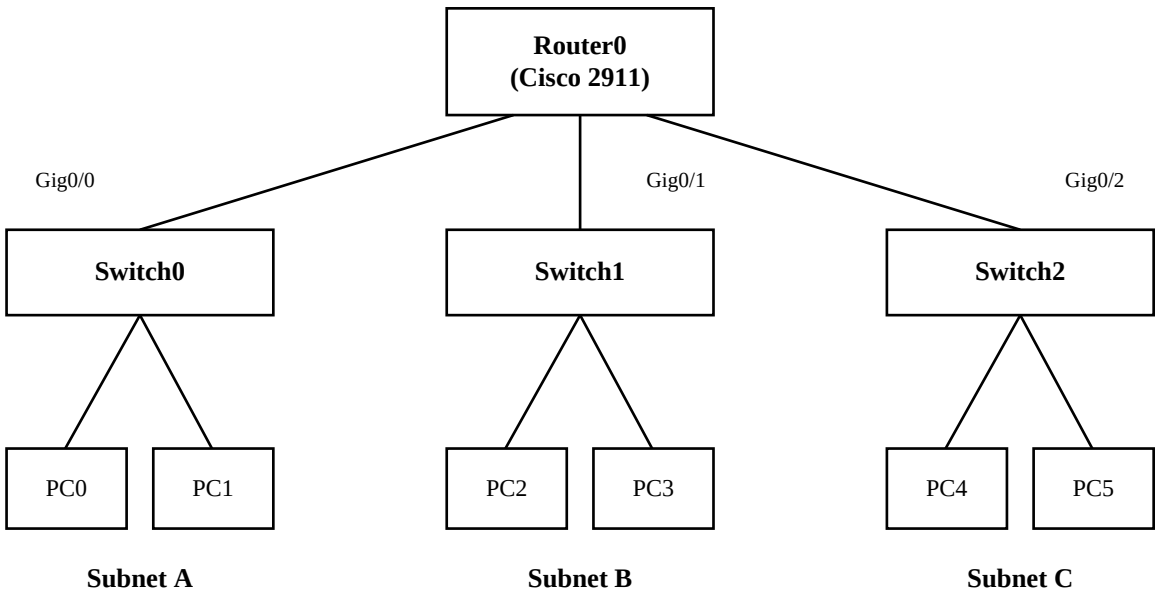
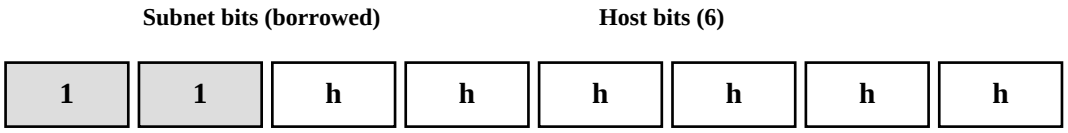


Fig 2: Borrowing 2 bits of the host part for subnetting



192.168.1.0 /24 -> /26 (2 bits borrowed = 4 subnets, 2^6 - 2 = 62 hosts each)

Subnet design:

Subnet	Network ID	Mask	Usable range	Broadcast	Gateway
A (Sales)	192.168.1.0/26	255.255.255.192	.1 - .62	192.168.1.63	192.168.1.1
B (HR)	192.168.1.64/26	255.255.255.192	.65 - .126	192.168.1.127	192.168.1.65
C (IT)	192.168.1.128/26	255.255.255.192	.129 - .190	192.168.1.191	192.168.1.129
Unused	192.168.1.192/26	255.255.255.192	.193 - .254	192.168.1.255	-

IP address table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Gig0/0	192.168.1.1	255.255.255.192	-
Router0	Gig0/1	192.168.1.65	255.255.255.192	-
Router0	Gig0/2	192.168.1.129	255.255.255.192	-
PC0	Fa0	192.168.1.2	255.255.255.192	192.168.1.1
PC1	Fa0	192.168.1.3	255.255.255.192	192.168.1.1
PC2	Fa0	192.168.1.66	255.255.255.192	192.168.1.65
PC3	Fa0	192.168.1.67	255.255.255.192	192.168.1.65
PC4	Fa0	192.168.1.130	255.255.255.192	192.168.1.129
PC5	Fa0	192.168.1.131	255.255.255.192	192.168.1.129

Router configuration:

```
Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#ip address 192.168.1.129 255.255.255.192
Router(config-if)#no shutdown
Router(config-if)#end
Router#write memory
```

OUTPUT:

Router0 CLI - show ip interface brief

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.1.65    YES manual up                    up
GigabitEthernet0/2     192.168.1.129   YES manual up                    up
```

PC0 Command Prompt - same subnet

```
PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

(a) When the subnets are connected via a router

PC0 Command Prompt - ping PC2 (Subnet B) through the router

```
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.66: bytes=32 time=3ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127
Reply from 192.168.1.66: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
```

PC0 Command Prompt - ping PC4 (Subnet C) through the router

```
PC>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time=2ms TTL=127
```

PC0 Command Prompt - ping PC4 (Subnet C) through the router (contd.)

```
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

(b) When the subnets are not connected (router removed)

PC0 Command Prompt - ping PC2 (Subnet B) without the router

```
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

PC0 Command Prompt - ping PC1 (same subnet) without the router

```
PC>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

OBSERVATION:

Case	Source -> Destination	Result
Router connected	PC0 -> PC1 (same subnet)	Success, TTL = 128
Router connected	PC0 -> PC2 / PC4 (other subnet)	Success, TTL = 127 (one router hop). First packet lost while ARP is resolved.
Router removed	PC0 -> PC1 (same subnet)	Success, switch forwards the frames
Router removed	PC0 -> PC2 / PC4 (other subnet)	Fails, 100% loss. No device can route between subnets.

RESULT:

Thus subnetting of the 192.168.1.0/24 network into /26 subnets was implemented in Cisco Packet Tracer. Hosts in different subnets could communicate only when a router was connected between them, while hosts inside the same subnet could communicate even without the router.